Adaptive traffic signal control based on Dueling Deep Q-Learning Network

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ABSTRACT

The intersection is the place where traffic flow converges and traffic accidents are most likely to occur, and the traffic conditions at intersections is also complex, which is easy to cause congestion and reduce the traffic efficiency. With the development of intelligent transportation system, adaptive traffic signal control is proposed to improve the traffic efficiency of the intersection by adjusting the phase sequence of the signal in real time. Meanwhile, with the extensive application of reinforcement learning in the traffic field, dueling DQN (Deep Q-Learning Network) is introduced to realize the adaptive traffic signal control in this paper. By comparing the traffic flow conditions of intersections under different signal control methods, we find that dueling DQN can effectively improve the traffic efficiency of intersections under the condition of random change of traffic flow. Through experiment results, we can conclude that the expected reward of dueling DQN is 41.1% and 4.8% higher than that of fixed phase method and DQN method. Meanwhile, simulation result shows that dueling DQN can react to the traffic flow of the road network more timely, and has higher stability than DQN, which can effectively improve the traffic efficiency.

CCS CONCEPTS

• **Machine learning** → Machine learning approaches; Neural networks.

KEYWORDS

Intelligent Transportation System, Reinforcement Learning, Dueling DQN, Adaptive Traffic Signal Control

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1 INTRODUCTION

With the development of economy and the increasing number of vehicles, the traffic around the world is becoming more and more crowded. Intersections are one of the places where traffic flows converge, which is easy to cause congestion, further leading to the increase of traffic accident rate. Therefore, how to solve intersection congestion and improve intersection traffic efficiency has become a research hotspot at home and abroad. With the development of big data, artistic intelligence, smart roads and vehicle road collaboration, China's intelligent transportation system is becoming mature [1–3]. Adaptive traffic signal control can adjust the traffic efficiency of intersections by adjusting the phase sequence of signal lights. Therefore, adaptive traffic signal control has become an important part of the intelligent transportation system, which has been studied by many scholars [4–6]. Azimirad E and others proposed a new fuzzy model and fuzzy logic controller for traffic light intersections. By using the state space equation, they established the average waiting time model of vehicles in the traffic network under fixed time control. By controlling the time and phase sequence of traffic lights, they can ensure smooth traffic with minimum waiting time and queue length [7]. Younes M B and others proposed an intelligent traffic light control algorithm based on the real-time traffic characteristics of competing traffic flows at traffic light intersections, which is capable to effectively arrange the phase of each traffic light, and then designed a traffic dispatching signal light control algorithm suitable for the main road scene to generate effective traffic scheduling for the entire road network [8]. Qi L and others designed a two-level control strategy for traffic light intersections to prevent urban traffic congestion caused by traffic accidents through additional traffic warning lights [9].

Recently reinforcement learning has the outstanding performance to solve complex problems [10], and the tremendous advancement of reinforcement learning, in various contexts, has allowed it to occupy an important place in the deep learning field [11]. Da Silva proposed a method called RL-CD (Reinforcement Learning with Context Detection), which performs well in the non-stationary

environment of intersections and is superior to the classical reinforcement learning algorithm Q-learning [12]. By using Bayesian interpretation and some innovative incentive designs, Khamis M A developed a multi-agent signal light control system based on the multi-objective sequential decision-making framework, which enhanced the matching between the control of a single signal controller and the environment [13]. Wang applied the proactive complex event processing method (in active traffic congestion control), which uses reinforcement learning to find the optimal joint strategy. When it is used to control congestion, the method works well [14]. Compared with traditional signal light control, reinforcement learning can select the phase sequence of signal lights according to the changing traffic flow, which can improve the traffic efficiency of intersections more effectively.

In this paper, we followed these achievements and experiences and realized the adaptive traffic signals control based on dueling DQN, comparing with different traffic signal control methods.

2 METHODS

With the rapid development of 5G, the data transmission speed has been greatly improved, and the communication between vehicles and road infrastructure has become more collaborative. The roadside infrastructure can obtain the traffic information of the road network in real time, providing a data basis for adaptive traffic signal control; meanwhile, with the development of intelligent transportation system and chip technology, the intelligence level of mobile edge computers will be greatly improved, and their information processing capacity will also be improved, providing the possibility for complex operations and timely control of the signal. Currently, with the rapid development of machine learning, reinforcement learning has been applied in a wide range of fields Therefore, this paper introduces reinforcement learning into the adaptive traffic signal control, by selecting the signal phase sequence, optimizing the control process and transmitting the signal to the signal controller through mobile edge computer processing.

2.1 Reinforcement Learning

The intersection is one of the places where traffic flow converges and traffic accidents are most likely to occur. Under the connected and automated environment, the process of the connected and automated vehicle passing through the intersection is more complex, which further leads to traffic congestion. Therefore, traffic signal control is considered to be one of the most critical issues in intelligent transportation systems. Adaptive traffic signal control is designed to adjust the phase sequence of the signal, according to the length of the vehicle queue at the intersection and the traffic flow on the road. Therefore, the current traffic flow and congestion of the current road network determine the process of adaptive traffic signal control, and the environmental characteristics are relatively discrete. Therefore, the traffic model and the signal phase sequence model of the intersection can be built through Markov chain for agents' learning and training.

Reinforcement learning is an important branch of machine learning, which usually includes four elements: policy, reward, value and environment. Reinforcement learning focuses on how to act based on the environment to maximize the expected benefits. That is, how the agent, stimulated by the reward or punishment given by the environment, gradually forms the expectation of stimulation and produces habitual behavior that can obtain the maximum benefit. Thus, reinforcement learning can transform intersection traffic flow model and traffic light model into learning process of agent, the following introduces how to use dueling DQN to perform adaptive traffic signal control.

2.2 Dueling DQN

Dueling DQN is developed from Q learning, which is a model free reinforcement learning method. Its principle is that the agent performs the action under the condition of obtaining the environment information and allocates Q value through evaluating the environment state, which can reflect the quality of the operation performed. Its main idea is to use ϵ-greedy strategy samples the data to generate the policy, and evaluates the action to update the Q value for each time step of the action. The traditional Q-learning environment is relatively simple and limited, which can list all the environment states, and each state has its corresponding optimal action and corresponding Q value. But for the complex intersection, when the queuing vehicles, traffic flow and signal phase sequence at the intersection are constantly changing, it is impossible to obtain the best behavior by searching the finite Q value table. However, neural networks can handle massive and complex data. Therefore, the DQN, which combines neural networks with Q-learning, is introduced to describe the intersection scene.

However, DQN still has the problem of slow learning speed, which is unable to quickly learn the changes of traffic flow and the phase sequence of signal signals. Therefore, dueling DQN is introduced to speed up the learning speed of agents and enable signals to respond to traffic flow more efficiently. Dueling DQN improves the DQN by establishing a new neural network structure – duel network, which effectively improves the performance of the algorithm. The network input of dueling DQN is the same as that of DQN, both of which are state information, but the output is different. The output of dueling DQN algorithm includes two branches, namely, the state value V (scalar) of the state and the advantage value A (vector with the same dimension as the action space) of each action. However, the output of DQN has only one branch. The network structure difference between dueling DQN and DQN is shown in figure 1

In the network structure of DQN, the convolutional network is used to extract the environment features, and then the action value of each action is output through the full connection layer, the Q value of DQN is only affected by the state. In the network structure of dueling DQN, after inputting the same environmental features, and then use the convolution network to extract the environmental features to obtain the feature vector. The output will go through two full connection layer branches, corresponding to the state value and dominant value respectively. Finally, add the state value and dominant value to obtain the action value of each action. Therefore, the dual network structure can include actions and states into the influencing factors of Q value at the same time, so that Q value can converge faster and the learning speed of the algorithm is accelerated. The Q value update process of dueling DQN is introduced as follows.

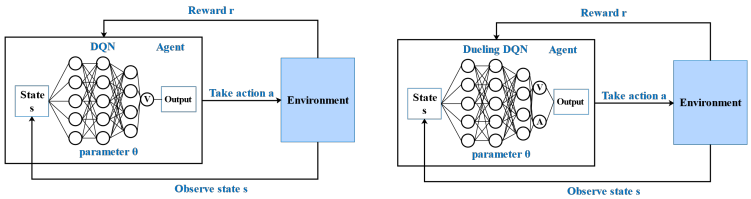


Figure 1: The network structure difference between DQN and Dueling DQN

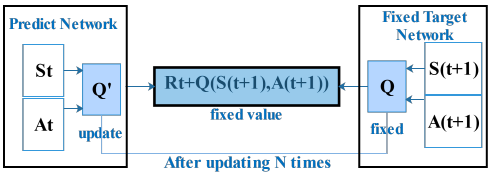


Figure 2: The fixed target network update process of dueling DQN

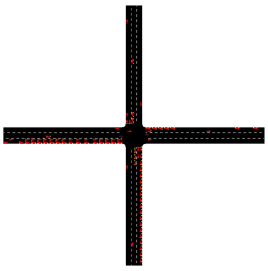


Figure 3: The intersection simulation scenario

Q-learning can obtain the maximum reward by evaluating the Q-value function of each action to select the action. Q is the expectation value that the action a (a ∈ A) can obtain under the s state (s ∈ S) at a certain time, and the environment will feedback the corresponding reward r according to the action of the agent. The following is the Q Value update formula

$$Q^*(S_t, A_t) = Q(S_t, A_t) + \alpha\left(R_{t+1} + \lambda \max Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\right) \quad (1)$$

Since every step after the current one will bring returns, the expected return is the sum of all returns ($\lambda$is the attenuation factor). The following is the Q value update function of dueling DQN

$$Q^*(s, a; \theta) = V(s; \theta) + A(s, a; \theta) \quad (2)$$

Where V() is the optimal state function, A() is the optimal advantage function and θ is the network parameter The two functions form the Q Value output of Dueling DQN. The following is the loss function of dueling DQN,

$$\mathrm{L}(\theta) = \mathrm{E}\left[(\mathrm{TargetQ} - \mathrm{Q}(\mathrm{s}, \mathrm{a}; \theta))^2\right] \quad (3)$$

The following is the target function of dueling DQN, where R is the update value and $\gamma$is a proportional parameter.,

$$\mathrm{TargetQ} = R + \gamma \max \mathrm{Q}(S_{t+1}, A_{t+1}; \theta) \quad (4)$$

The iterative process of Q-learning has the problem of unstable algorithm training. Since when the target network is updated, there will be conflicts between the input process and the update process, which will lead to unstable model training, easy to produce fluctuations, and the model convergence speed will be slower. In dueling DQN, two target models are established, one is the prediction model, and the other is the fixed target model. The prediction model updates the parameters in real time according to the input, while the fixed target model obtains the updated parameters from the prediction model after one cycle and remains unchanged in the next cycle. The fixed target network update process of dueling DQN is shown in figure 2.

3 SIMULATIONS

3.1 Simulation tool

SUMO (Simulation of urban mobility) is an open source and widely used traffic simulation software. Its car following model and lane change model are both reliable and have good algorithm compatibility, which is compatible with reinforcement learning algorithm to realize adaptive traffic signals adaptive control. This paper uses traci interface to simulate the situations at intersections in python.

3.2 Intersection simulation scenario

The simulation intersection scene in this paper is a two-way six lane intersection, which is divided into four directions: east, west, north and south. Each lane is 200 meters long, and each direction is composed of three lanes. In the simulation experiment, the traffic flow in four directions is set in a fixed interval, but the traffic flow in each direction is random. In this experiment, the influence of pedestrians on traffic flow and road network is not considered. The following figure 3 shows the intersection simulation scenario:

The following describes the setting of the state, action and reward of the intersection scenario. Dueling DQN can select the corresponding phase sequence of the signal lights by obtaining the state reward of the intersection to realize adaptive traffic signal control.

State: In this paper, the location and speed of vehicles at the intersection can be obtained through the grid division of roads, and then the location and speed of vehicles can be converted into the corresponding location matrix and speed matrix through matrix changes. Meanwhile, the program can periodically acquire the phase sequence state of the signal, so as to take the vehicle speed and position and the signal phase sequence as the state variables.

Action: In this paper, we set four groups of signal phase sequence combinations, which can be expressed as A∈{0, 1, 2, 3}. During the phase transition, this paper does not add additional transition

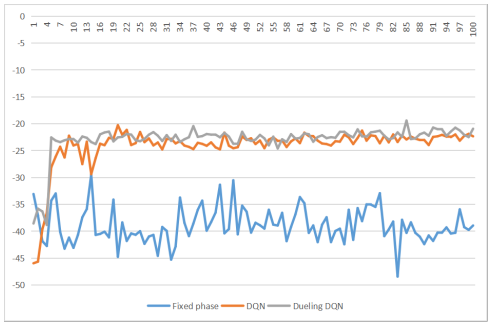


Figure 4: Expected reward in different signal control method

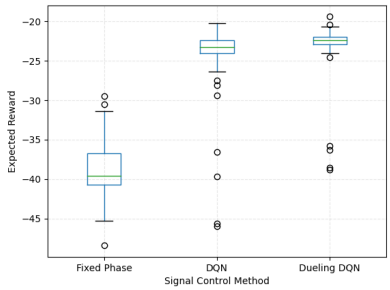


Figure 5: Average reward box chart in different signal control method

phases, but uses the yellow light delay to clear the intersection vehicles to prevent vehicle collisions.

Reward: The evaluation of the adaptive control effect of signal in this paper is mainly based on the traffic efficiency. Therefore, the model takes the waiting time at the intersection and the length of the queue as the reward of the system feedback after taking the action. The reward at time t is defined by the following equation:

$$R = w_1 \times \sum_{i \in I} N_i + w_2 \times \sum_{i \in I} W_i \tag{5}$$

where NI is the total number of vehicles in intersection with a speed less than 0.1m/s, Wi is the waiting time of all vehicles and I is the collection of all intersection vehicles. W1 and W2 are their respective weights.

4 RESULT AND CONCLUSION

After the simulation experiment, according to waiting time and queue length of vehicles, the expected reward under different signal control methods can be obtained. The figure 4 shows the expected reward obtained during simulation in different methods. It can be seen that the expected reward varies greatly in fixed phase method, which means that the changing traffic flow is easy to cause traffic jam with fixed signal phase sequence method. It can also be seen from the figure 4 that compared with fixed phase methods, dueling DQN and DQN have significantly improved their final convergence expected reward, indicating that they have significantly improved traffic efficiency. However, dueling DQN has higher convergence expected rewards than DQN; meanwhile, dueling DQN converges faster than the other two methods, which shows that agents of dueling DQN can perform adaptive traffic signal control more timely according to traffic flow.

The following figure 5 shows the expected reward box under different signal control methods. From the following box chart, we can get the average reward of dueling DQN, DQN and fixed phase method, which are -22.94, -24.09 and -38.90, indicating that the average reward interval of dueling DQN is higher than that of DQN and fixed phase, which are 4.8% and 41.1% respectively. Meanwhile, the expected average reward interval of dueling DQN is the most stable, which indicates that the changing traffic flow has little impact on adaptive traffic signal control based on dueling DQN.

In this paper, in order to improve the traffic efficiency of the intersection, dueling DQN is used for adaptive traffic signal control, so that the phase sequence of the signal can be changed in time when the traffic flow changes, and the traffic efficiency of the intersection can be improved. Through simulation and comparison of different signal control methods, dueling DQN can greatly improve the traffic efficiency of the road network compared with the traditional fixed phase method. Compared with DQN, dueling DQN has a faster learning speed and more stable expected reward, which makes the traffic condition at the intersection more stable and the vehicle and signal light are more integrated.

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